

Scientific Programming Lab - Unit 1

1. Program to convert temperatures to and from Celsius, Fahrenheit

Question: Program to convert temperatures to and from Celsius, Fahrenheit.

Aim: To convert temperature from Celsius to Fahrenheit and vice versa.

Objective: To understand input/output and basic arithmetic operations in Python.

Program:

```
def celsius_to_fahrenheit(c):  
    return (c * 9/5) + 32  
  
def fahrenheit_to_celsius(f):  
    return (f - 32) * 5/9  
  
c = float(input("Enter temperature in Celsius: "))  
print("Temperature in Fahrenheit:", celsius_to_fahrenheit(c))  
  
f = float(input("Enter temperature in Fahrenheit: "))  
print("Temperature in Celsius:", fahrenheit_to_celsius(f))
```

Result/Output:

```
Enter temperature in Celsius: 37  
Temperature in Fahrenheit: 98.6  
Enter temperature in Fahrenheit: 98.6  
Temperature in Celsius: 37.0
```

Result Analysis: The program gives the desired output according to the objective successfully.

2. Program to perform different Arithmetic Operations on numbers in Python

Question: Program to perform different Arithmetic Operations on numbers in Python.

Aim: To perform addition, subtraction, multiplication, division, and modulus.

Objective: To apply arithmetic operators in Python.

Program:

```
a = float(input("Enter first number: "))  
b = float(input("Enter second number: "))  
print("Addition:", a + b)
```

```
print("Subtraction:", a - b)
print("Multiplication:", a * b)
print("Division:", a / b)
print("Modulus:", a % b)
```

Result/Output:

Enter first number: 10

Enter second number: 3

Addition: 13.0

Subtraction: 7.0

Multiplication: 30.0

Division: 3.3333333333333335

Modulus: 1.0

Result Analysis: The program gives the desired output according to the objective successfully.

3. Write a python program to check if a given number is even or odd

Question: Write a python program to check if a given number is even or odd.

Aim: To determine if a number is even or odd using modulus.

Objective: To apply conditional statements in Python.

Program:

```
num = int(input("Enter a number: "))
if num % 2 == 0:
    print("Even Number")
else:
    print("Odd Number")
```

Result/Output:

Enter a number: 7

Odd Number

Result Analysis: The program gives the desired output according to the objective successfully.

4. Write a python program to find largest of three numbers

Question: Write a python program to find largest of three numbers.

Aim: To compare three numbers and find the maximum.

Objective: To use decision-making statements effectively.

Program:

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
c = int(input("Enter third number: "))

if a >= b and a >= c:
    print("Largest number is:", a)
elif b >= a and b >= c:
    print("Largest number is:", b)
else:
    print("Largest number is:", c)
```

Result/Output:

```
Enter first number: 12
Enter second number: 25
Enter third number: 7
Largest number is: 25
```

Result Analysis: The program gives the desired output according to the objective successfully.

5. Program to print overall grade of a student in python

Question: Program to print overall grade of a student in python.

Aim: To calculate grade based on percentage.

Objective: To implement conditional logic for grading.

Program:

```
marks = float(input("Enter your percentage: "))

if marks >= 90:
    print("Grade: A")
elif marks >= 80:
    print("Grade: B")
elif marks >= 70:
    print("Grade: C")
elif marks >= 60:
    print("Grade: D")
```

else:

```
    print("Grade: F")
```

Result/Output:

Enter your percentage: 86

Grade: B

Result Analysis: The program gives the desired output according to the objective successfully.

6. Program to print 3 numbers in ascending order

Question: Program to print 3 numbers in ascending order.

Aim: To sort and display three numbers.

Objective: To practice list sorting.

Program:

```
nums = []  
  
for i in range(3):  
    nums.append(int(input("Enter number: ")))  
  
nums.sort()  
  
print("Ascending order:", nums)
```

Result/Output:

Enter number: 25

Enter number: 10

Enter number: 15

Ascending order: [10, 15, 25]

Result Analysis: The program gives the desired output according to the objective successfully.

7. Program to find factorial of a given number

Question: Program to find factorial of a given number.

Aim: To calculate factorial using a loop.

Objective: To understand loops and multiplication operations.

Program:

```
num = int(input("Enter a number: "))  
  
factorial = 1  
  
for i in range(1, num + 1):  
    factorial *= i
```

```
print("Factorial:", factorial)
```

Result/Output:

Enter a number: 5

Factorial: 120

Result Analysis: The program gives the desired output according to the objective successfully.

8. Program to print natural numbers and find the sum up to given limit

Question: Program to print natural numbers and find the sum up to given limit.

Aim: To print numbers from 1 to n and calculate their sum.

Objective: To use loops and summation logic.

Program:

```
n = int(input("Enter limit: "))
```

```
total = 0
```

```
for i in range(1, n + 1):
```

```
    print(i, end=" ")
```

```
    total += i
```

```
print("\nSum:", total)
```

Result/Output:

Enter limit: 5

1 2 3 4 5

Sum: 15

Result Analysis: The program gives the desired output according to the objective successfully.

9. Program to print Fibonacci series up to a given limit

Question: Program to print Fibonacci series up to a given limit.

Aim: To generate the Fibonacci series.

Objective: To apply loops and recursion logic.

Program:

```
n = int(input("Enter limit: "))
```

```
a, b = 0, 1
```

```
while a <= n:
```

```
    print(a, end=" ")
```

```
    a, b = b, a + b
```

Result/Output:

Enter limit: 15

0 1 1 2 3 5 8 13

Result Analysis: The program gives the desired output according to the objective successfully.

10. Print multiplication table of a given number

Question: Print multiplication table of a given number.

Aim: To display the multiplication table of a single number.

Objective: To practice loops and multiplication.

Program:

```
n = int(input("Enter number: "))  
for i in range(1, 11):  
    print(f"{n} x {i} = {n*i}")
```

Result/Output:

Enter number: 5

5 x 1 = 5

...

5 x 10 = 50

Result Analysis: The program gives the desired output according to the objective successfully.

11. Print multiplication tables up to given limit

Question: Print multiplication tables up to given limit.

Aim: To generate tables for multiple numbers.

Objective: To use nested loops effectively.

Program:

```
limit = int(input("Enter the limit: "))  
for n in range(1, limit + 1):  
    print(f"Table for {n}")  
    for i in range(1, 11):  
        print(f"{n} x {i} = {n*i}")  
    print()
```

Result/Output:

Enter the limit: 3

Table for 1

$1 \times 1 = 1$

...

Table for 3

$3 \times 10 = 30$

Result Analysis: The program gives the desired output according to the objective successfully.

12. Program to check if a given number is prime or not

Question: Program to check if a given number is prime or not.

Aim: To check if a number has only two factors: 1 and itself.

Objective: To apply loops and conditional statements.

Program:

```
num = int(input("Enter a number: "))
if num > 1:
    for i in range(2, int(num**0.5) + 1):
        if num % i == 0:
            print(num, "is not a prime number")
            break
    else:
        print(num, "is a prime number")
else:
    print(num, "is not a prime number")
```

Result/Output:

Enter a number: 13

13 is a prime number

Result Analysis: The program gives the desired output according to the objective successfully.